

BioSig for Biomedical Signal Processing

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In this article we discuss BioSig library, an open source software for biomedical signal processing. The library provides open source codes for various application areas including neuroinformatics, brain-computer interfaces, neurophysiology, psychology, cardiovascular systems and sleep research. One could effectively use BioSig library for the analysis of various bio-signals like electrocardiogram (ECG), electroencephalogram (EEG), electrocorticogram (ECoG), respiratory signals and more.

Check out the latest release of this software, released under GNU general-public licence, bundled in this month's EFY Plus DVD.

In quest of a biomedical signal processing tool

While dealing with biomedical signals, there are a lot of challenges that need to be addressed. One is the difference in data format for the storage of signals. Low value of signal-to-noise ratio, artefact contamination and choice of apt data processing techniques for various signal modalities are some others. Again, development of the software is by itself an integral part of biomedical signal processing.

Even though MATLAB is capable of processing biomedical signals, algorithms are not openly accessible. This leads to re-implementation, which is essentially a waste of time, money and effort.

Compatibility issues rising from the proprietary solutions offered by the providers for the software development is another issue. There is no unique data format for the storage of every bio-signal. The customer has to depend on the vendor's proprietary software. Also, there are

What the package includes

The BioSig package contains a handful of useful libraries for various software. The main libraries are:

bioSig4octmat. This toolbox contains features including essential feature-extraction algorithms, classification methods, power filters for data import and export and a powerful viewing and scoring software. Even though this library is intended for use in MATLAB and Octave, use of the toolbox in the latter is somewhat slower.

bioSig4c++. This library enables the user to deal with bio-signal data formats from C/C++.

rtSBCI. The BioSig package is capable of handling BCI applications, too. The rtSBCI library provides the user with a framework for the rapid prototyping and development of real-time BCI systems.

Bioprofeed. The bioprofeed tool helps in the acquisition, processing, storage and feedback generation of biomedical data. Main components in the toolbox are transducers, a biosignal amplifier, an ADC and a processing and storage unit.

Bcix. This is yet another interesting software package for dealing with BCI applications.

possibilities of loss of data while it is being exported.

All these factors were collectively responsible for the emergence of BioSig as a novel biomedical signal-processing software.

Where BioSig could be used

The research article titled 'BioSig: The Free and Open Source Software Library for Biomedical Signal Processing' by Carmen Vidaurre, Tilmann H. Sander and Alois Schlögl discusses some areas where the software is actively deployed.

Extracting heart rate from ECG.

The toolbox contains two algorithms that help in the extraction of heart rates in a single routine. The heart rates so computed can be efficiently used for the advanced studies of heart-rate variability.

In the first algorithm, a Hilbert transform is used to determine the envelope of ECG signals, threshold values are set and corresponding R-peak positions are determined. The second algorithm makes use of filter banks and incorporates an ectopic

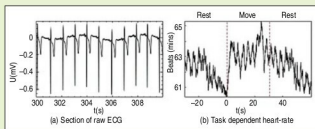


Fig. 1: BioSig has algorithms capable of extracting heart-rate from an ECG signal

beat correction.

Artefact processing. BioSig includes several artefact-processing techniques in its library. An algorithm helps in the detection of muscular noise in an awake EEG, implemented using time-domain and frequency-domain methods.

For biomedical signals, the software also has a quality-control method based on histogram analysis. This method helps in the visual identification of the thresholds needed for saturation detection.

Yet another artefact-removal technique available in the library is for the removal of electrooculographic (EOG) artefacts in an EEG based on simple linear regression.

Fostering the studies in neuroscience. Multiple areas in the human brain interact with each other while